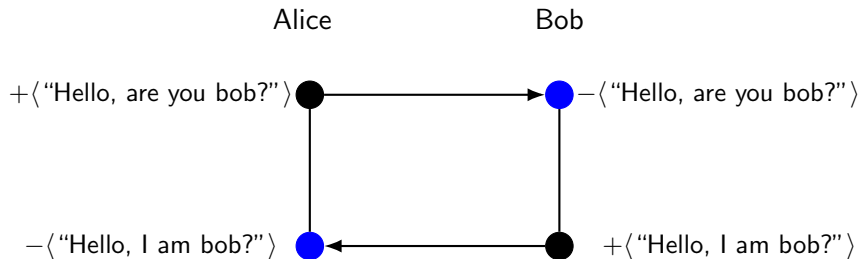


Session 3: CPSA Etymology and Attacks in CPSA

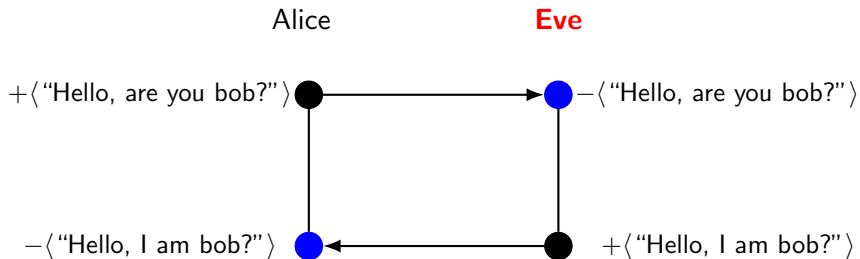
Slides prepared by UMBC Protocol Analysis Lab

Last edited May 28, 2026

A completed protocol?



Strands can lie



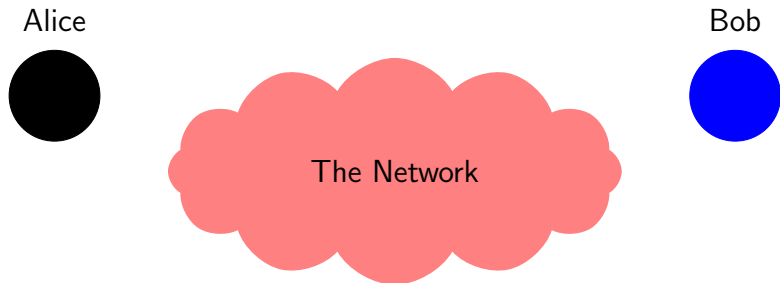
CPSA adversarial strand actions

- M. Text Message: $\langle +t \rangle$ where $t \in T$
- T. Tee: $\langle -g, +g, +g \rangle$ where $g \in G$
- C. Concatenation: $\langle -g, -h, +gh \rangle$
where $g, h, gh \in G$
- S. Separation into components:
 $\langle -gh, +g, +h \rangle$
- K. Key: $\langle +K \rangle$ where $K \in \mathbf{K}_p$
- E. Encryption $\langle -K, -h, +\{h\}_K \rangle$
- D. Decryption $\langle -K^{-1}, -\{h\}_K, +h \rangle$

1. $\langle +m \rangle$ emit a message from the network.
2. $\langle -m \rangle$ extract a message from the network.
3. $\langle +m \rangle$ concatenate two messages.
4. $\langle +m \rangle$ encrypt a message m under key K .

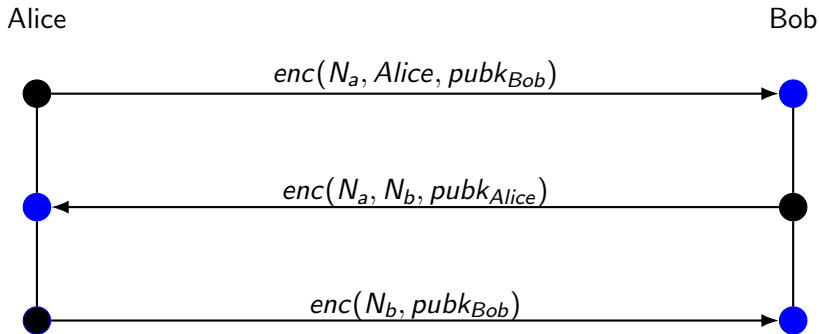
An adversarial model of the network

An adversarial model of the network



Goal: Mutually *authenticate* (verify identities) over an insecure network.

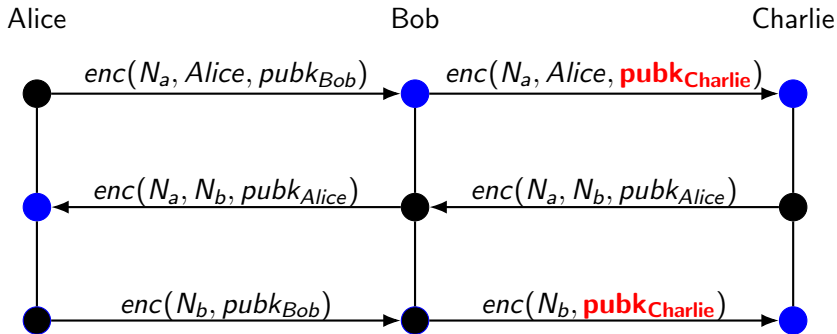
Needham Schroeder



Is this protocol secure against the Dolev-Yao Adversary?

● = Reception ● = Transmission

Needham Schroeder



Remember: The Dolev-Yao Adversary can be a legitimate party on the network

Blue circle = Reception Black circle = Transmission

What do we give CPSA?

`protocol.scm`

What do we give CPSA?

`protocol.scm`

▶ `herald`

What do we give CPSA?

`protocol.scm`

- ▶ `herald`
- ▶ `defmacro`

What do we give CPSA?

`protocol.scm`

- ▶ `herald`
- ▶ `defmacro`
- ▶ `defprotocol`

What do we give CPSA?

`protocol.scm`

- ▶ `herald`
- ▶ `defmacro`
- ▶ `defprotocol`
- ▶ `defskeleton`
- ▶ `defstrand`

What does CPSA give us?

`protocol.scm` →

What does CPSA give us?

`protocol.scm` →

▶ `protocol.txt`

What does CPSA give us?

`protocol.scm` →

- ▶ `protocol.txt`
- ▶ `protocol_shapes.txt`

What does CPSA give us?

`protocol.scm` →

- ▶ `protocol.txt`
- ▶ `protocol_shapes.txt`
- ▶ `protocol.xhtml`

What does CPSA give us?

`protocol.scm` →

- ▶ `protocol.txt`
- ▶ `protocol_shapes.txt`
- ▶ `protocol.xhtml`
- ▶ `protocol_shapes.xhtml`

How can we spot attacks?

How can we spot attacks?

► *SECURITY GOALS*

How can we spot attacks?

- ▶ *SECURITY GOALS*
- ▶ Red circles

How can we spot attacks?

- ▶ *SECURITY GOALS*
- ▶ Red circles
- ▶ Dotted lines

How can we spot attacks?

- ▶ *SECURITY GOALS*
- ▶ Red circles
- ▶ Dotted lines
- ▶ Listener Strands

Homework

Find attacks in CPSA

Find attacks in CPSA

Looking at shapes to find attacks takes some practice, so let us do some practice!

Three of the protocols are already done, you just have to run `make` so CPSA can make shapes, and then analyze them.

Two of the protocols need to be filled in, you will have to add part of a role, fill in a skeleton, or something else (documented in the comments and README), and then run CPSA to analyze the shapes.

Survey

Fill in details about the survey here!